



DEPARTAMENTO DE SEGURIDAD Y DEFENSA

OBJECT-ORIENTED PROGRAMMING

THE SOFT WARRIORS

Project description

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Project Overview

The Store Management System is a software solution designed to support the daily operations of a retail store. The system helps store managers and cashiers manage sales, backups, cash control, customer credits, promotional combos, expiration alerts, and product code administration.

The project was developed to improve operational efficiency, reduce manual processes, and provide better control over inventory, transactions, and business data.

Problem Statement

Many small and medium-sized stores manage their operations manually or with limited software tools. This can lead to problems such as:

- 1) Incorrect cash balance calculations
- 2) Loss of important backup information
- 3) Difficulties managing promotional combos
- 4) Duplicate product codes
- 5) Poor expiration date tracking
- 6) Delays in customer credit management
- 7) Lack of operational organization

The system was created to solve these issues through a centralized and organized management platform.

General Objective

Develop a store management system capable of improving inventory control, sales management, customer credit administration, and operational efficiency.

Specific Objectives

- Implement a backup management module
- Create cash control functionalities
- Manage promotional combo products
- Handle customer credit payments and limits
- Generate expiration alerts for products
- Manage multiple product codes
- Improve data organization and validation
- Reduce operational errors during sales processes

Main Features

1. **Combo Management System**
Allows the store manager to create promotional bundles and manage combo activation and product assignment.
2. **Reservation Management**
Allows products to be reserved for customers while updating the available stock automatically.
3. **Returnable Container Tracking**
Tracks returnable containers between customers, suppliers, and the store during sales transactions.
4. **Critical Stock Control**
Calculates minimum stock levels and generates suggested supplier order lists before products run out.
5. **Supplier Invoice Management**
Registers supplier invoices, detects cost increases, and suggests updated retail prices.
6. **Expiration Control System**
Generates expiration alerts and manages notifications for products close to expiration dates.
7. **Customer Loyalty Recovery**
Identifies inactive customers and suggests personalized offers to recover customer loyalty.
8. **Credit Management System**
Manages customer debts, payments, credit limits, and account restrictions.
9. **Cross-Selling Recommendation System**
Suggests complementary products during sales transactions based on purchasing habits.
10. **Tax and Fiscal Summary Management**
Classifies sales by tax category and generates monthly fiscal summaries for accounting purposes.
11. **Cash Control System**
Allows cashiers to manage the cash register, calculate totals, register withdrawals, and verify balance differences.
12. **Sales Transaction Management**
Manages sales records including products, payment types, suspended sales, and transaction processing.

Expected Benefits

- Improved organization of store operations
- Faster sales and transaction management
- Better backup and data recovery processes
- Improved inventory and expiration monitoring
- Reduction of manual calculation errors

Conclusion

The Store Management System provides a practical solution for managing essential store operations through a modular and organized software structure. The project focuses on improving reliability, operational control, and efficiency while supporting future scalability and additional features.